

UST- KHAIRYUZOVO, Russian Federation

WMO: 322870

Lat: 57.095N

Lon: 156.738E

Elev: 26

StdP: 14.68

Time Zone: 12.00 (E12)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.3	-14.5	-24.2	1.4	-18.9	-19.2	1.9	-13.9	27.7	16.4	26.2	17.0	5.8	130	0.809

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	67.0	60.7	63.9	58.5	61.4	57.0	61.8	65.3	59.5	62.5	57.8	60.3	10.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.3	78.2	63.2	58.1	72.1	61.0	56.4	67.7	59.3	27.8	65.9	26.2	63.0	25.0	60.5	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.4	22.1	19.5		DB	-26.2	72.5	4.0	3.0	-29.0	74.7	-31.4	76.5	-33.6	78.2	-36.5	80.4
				WB	-26.4	65.6	3.9	2.7	-29.2	67.6	-31.5	69.2	-33.7	70.8	-36.6	72.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.7	8.7	8.2	16.6	27.6	38.5	47.2	54.5	55.2	47.2	36.5	24.7	14.3	
	DBStd	18.52	12.51	12.57	11.10	7.82	4.82	4.62	4.29	4.19	4.78	5.92	8.45	11.12	
	HDD50	7034	1279	1171	1034	673	359	105	8	9	110	418	760	1108	
	HDD65	12145	1744	1591	1499	1123	822	533	327	305	535	883	1210	1573	
	CDD50	366	0	0	0	0	2	23	147	169	25	0	0	0	
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH74	2	0	0	0	0	0	0	2	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	8.8	8.6	8.3	8.8	9.5	9.4	9.0	8.2	8.9	8.5	8.5	9.1	8.5	
Precipitation	PrecAvg	21.9	0.9	0.6	0.6	0.9	1.3	1.8	2.5	3.1	2.8	3.1	2.7	1.6	
	PrecMax	34.7	2.4	2.2	1.9	2.0	3.2	5.7	6.8	6.7	7.9	7.8	5.4	5.0	
	PrecMin	13.2	0.1	0.0	0.0	0.2	0.2	0.0	0.1	0.3	0.4	0.5	1.0	0.2	
	PrecStd	5.4	0.6	0.5	0.4	0.5	0.7	1.3	1.8	1.8	1.6	1.5	1.2	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	39.7	41.2	49.1	60.0	68.7	72.6	69.7	62.9	54.1	44.8	43.0	
		MCWB	31.7	33.3	34.4	40.2	49.5	55.1	62.1	63.1	57.6	45.8	38.9	38.3	
	2%	DB	34.5	35.4	37.3	44.0	54.3	61.5	67.4	66.7	58.7	50.2	40.7	36.7	
		MCWB	29.4	30.5	31.3	37.4	46.0	53.0	60.8	61.6	54.1	44.5	36.0	31.7	
	5%	DB	31.4	32.2	34.4	40.5	50.6	58.2	64.2	63.9	56.5	47.9	37.7	33.3	
		MCWB	27.4	27.8	29.8	35.4	44.0	51.6	58.6	59.5	53.0	43.6	33.9	29.1	
	10%	DB	27.4	27.0	31.3	37.4	46.8	55.5	61.6	61.5	54.6	45.3	34.9	29.6	
		MCWB	24.6	24.5	27.9	33.6	42.0	50.0	57.3	57.8	51.7	41.9	32.3	27.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	33.5	34.6	41.9	50.6	56.8	65.0	65.6	59.9	48.4	39.9	38.3
			MCDB	36.9	38.6	40.8	46.4	57.5	63.1	67.3	67.6	61.4	51.1	42.3	42.9
2%		WB	29.6	30.7	32.0	38.2	46.8	54.1	61.7	62.6	56.0	46.1	36.4	32.4	
		MCDB	33.8	35.4	36.4	42.9	52.7	59.7	66.1	65.5	57.5	48.9	40.1	36.2	
5%		WB	27.6	27.8	30.0	36.1	44.4	52.2	59.5	60.3	53.7	44.2	34.3	29.4	
		MCDB	31.4	31.5	33.9	39.9	49.8	56.7	63.2	62.9	55.5	46.7	36.8	32.8	
10%		WB	24.6	24.5	27.9	33.9	42.3	50.6	57.7	58.4	52.1	42.4	32.4	26.8	
		MCDB	27.3	27.2	31.2	36.9	46.6	54.6	61.0	60.4	54.1	45.3	34.7	29.7	
Mean Daily Temperature Range		5% DB	MDBR	11.9	13.5	13.8	11.3	11.3	11.3	10.4	9.7	11.6	9.4	9.1	12.0
			MCDBR	12.4	12.8	13.1	14.8	18.8	17.7	15.2	14.4	13.2	13.3	10.6	11.5
	MCWBR		9.8	9.7	9.6	9.8	11.8	10.4	9.4	8.9	9.6	9.4	7.8	9.1	
	5% WB	MCDBR	12.3	12.7	12.4	12.9	17.4	15.0	13.5	12.0	10.8	11.5	9.9	11.4	
		MCWBR	9.7	10.3	9.4	8.9	11.6	10.1	9.2	8.6	9.0	9.3	7.8	9.3	
Clear-Sky Solar Irradiance	taub	0.261	0.298	0.344	0.425	0.467	0.436	0.455	0.427	0.361	0.349	0.301	0.264		
	taud	2.134	2.107	2.030	1.898	1.958	2.170	2.150	2.227	2.395	2.241	2.117	2.052		
	Ebn at Noon	192	232	250	245	243	255	246	245	249	211	167	148		
	Edh at Noon	17	28	39	52	53	43	43	38	28	24	18	14		
All-Sky Solar Radiation	RadAvg	180	449	920	1314	1571	1631	1526	1105	818	438	174	105		
	RadStd	18	40	89	194	166	202	157	125	77	44	21	11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page